

Duration: 3hrs

[Max Marks:80]

- N.B.: (1) Question No 1 is Compulsory.
(2) Attempt any three questions out of the remaining five.
(3) All questions carry equal marks.
(4) Assume suitable data, if required and state it clearly.

- 1 Attempt any **FOUR** **[20]**
 - a What are the two primary types of XML parsers?
 - b Explain the steps involved in establishing a JDBC connection to a database.
 - c What are the benefits of using jQuery for AJAX?
 - d Explain JSON.parse() and JSON.stringify() methods with example
 - e Write a code to display "Hello World" in React JS.
- 2 a Compare and contrast the traditional web approach with the AJAX approach. **[10]**
b Write a PHP application that includes a form for user input and processes the submitted data. **[10]**
- 3 a Explain the concept of drag-and-drop functionality in HTML5 with example. **[10]**
b Discuss the JSP lifecycle and its phases. **[10]**
- 4 a Explain different ways to apply CSS styles to HTML elements with example? **[10]**
b Write JavaScript code to check mobile number (mobile number should start with 9 or 8) and mail id (check @ and .). **[10]**
- 5 a Explain the concept of cookies and their use in session management. **[10]**
b Explain the following document object properties with proper syntax? **[10]**
 - i. document.getElementById()
 - ii. document.getElementsByClassName()
 - iii. document.getElementsByName()
- 6 a Describe the process of sending an HTTP request from a web client to a server and receiving a response. **[10]**
b Write a short note on JSX. **[10]**

Time: 3 Hours

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- N.B. (1) Question one is Compulsory.
 (2) Attempt any 3 questions out of the remaining.
 (3) Assume suitable data if required.

Q. 1 Solve any Four out of the following (5 marks each) 20

- Explain the CMM model.
- Explain the Requirements model.
- Explain the LOC.
- Different between Alpha and Beta testing
- Discuss the different level of DFD.

Q.2 a Explain Risk and its types? Explain the steps involved in setting up or generating RMMM plan. 10

b. Explain the Spiral model of software development 10

Q. 3 a) Explain the general format of SRS for Hospital Management system. 10

b) Explain the FP Estimation techniques in details. 10

Q. 4 a) Explain cohesion and Coupling. Explain different types with detailed example. 10

b) Explain the different techniques in white box testing. 10

Q. 5 a) Explain steps in version and change control. 10

b) Explain software Re- engineering in detail. 10

Q. 6 Solve any Four 20

- Compare FTR and Walkthrough
- What are the different types of maintenance?
- Explain the tracking and scheduling.
- Explain the Use Case Diagram.
- Different between White box and Block box Testing.

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(3) All questions carry equal marks.
(4) Assume suitable data, if required and state it clearly.

- 1 Attempt any FOUR [20]**
- A** Explain with examples the classification of IPv4 addresses. **5**
 - B** Create 7-bit hamming code for the message bit 1110 with even parity. **5**
 - C** List the advantages of fiber optics as a communication Medium. **5**
 - D** What are sockets and its different types? **5**
 - E** What is the use of DNS in networking? **5**
- 2 Attempt the following**
- A** For the class C network 193.160.10.0 having subnet mask 255.255.255.192 finds the number of subnet created and Number of host per subnet. **10**
 - B** What is network traffic congestion? How congestion is controlled? **10**
- 3 Attempt the following**
- A** Explain different framing methods? What is the advantage of variable length frame over fixed layer frame? **10**
 - B** How collision is controlled at MAC layer using CSMA/CD? **10**
- 4 Attempt the following**
- A** What is use of ARP protocol? And also discuss header structure for ARP packet. **10**
 - B** Which algorithms are useful for dynamic routing? **10**
- 5 Attempt the following**
- A** Explain the process of connection management at TCP layer. **10**
 - B** With the help TCP segment header structure, discuss importance of sequencing. **10**
- 6 Attempt the following**
- A** Compare FTP and Telnet. **10**
 - B** What are the benefits of Selective repeat ARQ protocol over Go-back N ARQ protocol? **10**

Time: 3 hours

Total Marks: 80

Note: 1. Question no.1 is compulsory.

- 2. Attempt any three out of remaining five.**
- 3. Assumptions made should be clearly indicated.**
- 4. Figures to the right indicates full marks.**
- 5. Assume suitable data whenever necessary.**

Q. 1 Solve any four. (05 marks each)

- A Every data structure in the data warehouse contains the time element. Why?
- B Explain FP Growth Algorithm.
- C Explain different types of attributes.
- D Discuss different applications of Web Mining.
- E Explain Holdout and Random subsampling method to evaluate the accuracy of classifier.
- F Differentiate between Classification and Clustering.

Q.2 (10 marks each)

- A For a supermarket chain, consider the following dimensions namely product, store, time and promotion. The schema contains a central fact table for sales with three measures unit_sales, dollars_sales and dollar_cost.

1. Draw a star schema.
2. Calculate the maximum number of base fact table records for warehouse with the following values given below:
 - Time period 5 years
 - Store-300 stores reporting daily sales
 - Product-40,000 products in each store (about 4000 sell in each store daily)
 - Promotion- a sold item may be in only one promotion in a store on a given day.

- B Explain the different techniques to handle noisy data.

Suppose a group of sales price records has been sorted as follows:

3, 7, 8, 13, 22, 22, 22, 26, 26, 28, 30, 37.

Partition them into three bins by equal-frequency (Equi-depth) partitioning method. Perform data smoothing by bin mean and bin boundary.

Q.3

(10 marks each)

- A Explain Updates to dimensional table in detail.
- B Explain the following data pre-processing methods.
I) Dimensionality reduction II) Data transformation and Discretization

Q.4

(10 marks each)

- A Given the training data for height classification, classify the tuple,
 $t = \langle \text{Rohit, M, 1.95} \rangle$ using Naïve Bayes Classification.

Name	Gender	Height	Output
Kiran	F	1.6m	Short
Jatin	M	2m	Tall
Madhuri	F	1.09m	Medium
Manisha	F	1.88m	Medium
Shilpa	F	1.7m	Short
Bobby	M	1.85m	Medium
Kavita	F	1.6m	Short
Dinesh	M	1.7m	Short
Rahul	M	2.2m	Tall
Shree	M	2.1m	Tall
divya	F	1.8m	Medium
Tushar	M	1.95m	Medium
Kim	F	1.9m	Medium
Aarti	F	1.8m	Medium
Rajashree	F	1.75m	Medium

- B Consider four objects with two attribute (X and Y). These four objects are to be grouped together into two clusters using k-means clustering algorithm. Following are the objects with their attribute values.

Object	X	Y
A	1	1
B	2	1
C	4	3
D	5	4

Q. 5

(10 marks each)

- A Given the following data, apply the Apriori algorithm. Find frequent item set and strong association rules. Given Support threshold=50%, Confidence=60%

Transaction	Items
T1	I1, I2, I3
T2	I2, I3, I4
T3	I4, I5
T4	I1, I2, I4
T5	I1, I2, I3, I5
T6	I1, I2, I3, I4

- B What is Web Mining? Differentiate between Web Mining and Data Mining. Explain types of Web Mining.

Q. 6

Write short note on.

(5 marks each)

- A Decision Tree Induction Algorithm
 B K-medoids clustering Algorithm
 C Multilevel and multidimensional association rule mining
 D Page Rank Algorithm

(3 hours)

[80 marks]

NOTE:

1. Question No 1 is compulsory.
2. Attempt any three questions from remaining.
3. Assume suitable data if necessary and state the same.

- Q1 Solve all questions below 20
- a) Design a Mealy Machine to identify if string starts with prefix ab over $\{a,b\}^*$.
 - b) Construct a DFA for accepting all strings over $\{a,b\}$ with substring abb.
 - c) Explain Universal Turing Machine.
 - d) Compare DFA and PDA
- Q2
- a) Find regular expression (RE) for all strings starting with b and ending in ba over $\{a,b\}$. Design NFA with epsilon moves for this RE. Convert it to equivalent DFA. 10
 - b) Find the Context Free Grammar for following 10
 - i. $L = \{a^i b^j c^k \mid i = j + k\}$
 - ii. $L = \{x \in \{0,1\}^* \mid x \text{ has equal number of zeros and ones}\}$
- Q3
- a) Convert following grammar to Chomsky Normal Form 10
 $S \rightarrow AACD, A \rightarrow aAb \mid \epsilon, C \rightarrow aC \mid a, D \rightarrow aDa \mid bDb \mid \epsilon.$
 - b) State closure properties of Regular languages and Context Free Languages. 10
- Q4
- a) Design PDA for $\{a^n x^n \mid n \geq 0, x \in \{b\}^*\}$. Comment if it is deterministic or not. 10
 - b) Find minimum state Finite Automata accepting $(01^*0 + 10^*)$. First design a NFA with epsilon moves. 10
- Q5
- a) State pumping lemma for context-free-languages. Apply pumping lemma to $L = \{ss \mid s \in \{a,b\}^*\}$. 10
 - b) Design a Turing Machine to add two unary numbers. Show simulation of the machine. 10
- Q6 Write Detailed note on (Any two) 20
- a) Applications of FA, PDA and TM.
 - b) Types of Turing Machines.
 - c) Chomsky Hierarchy.
